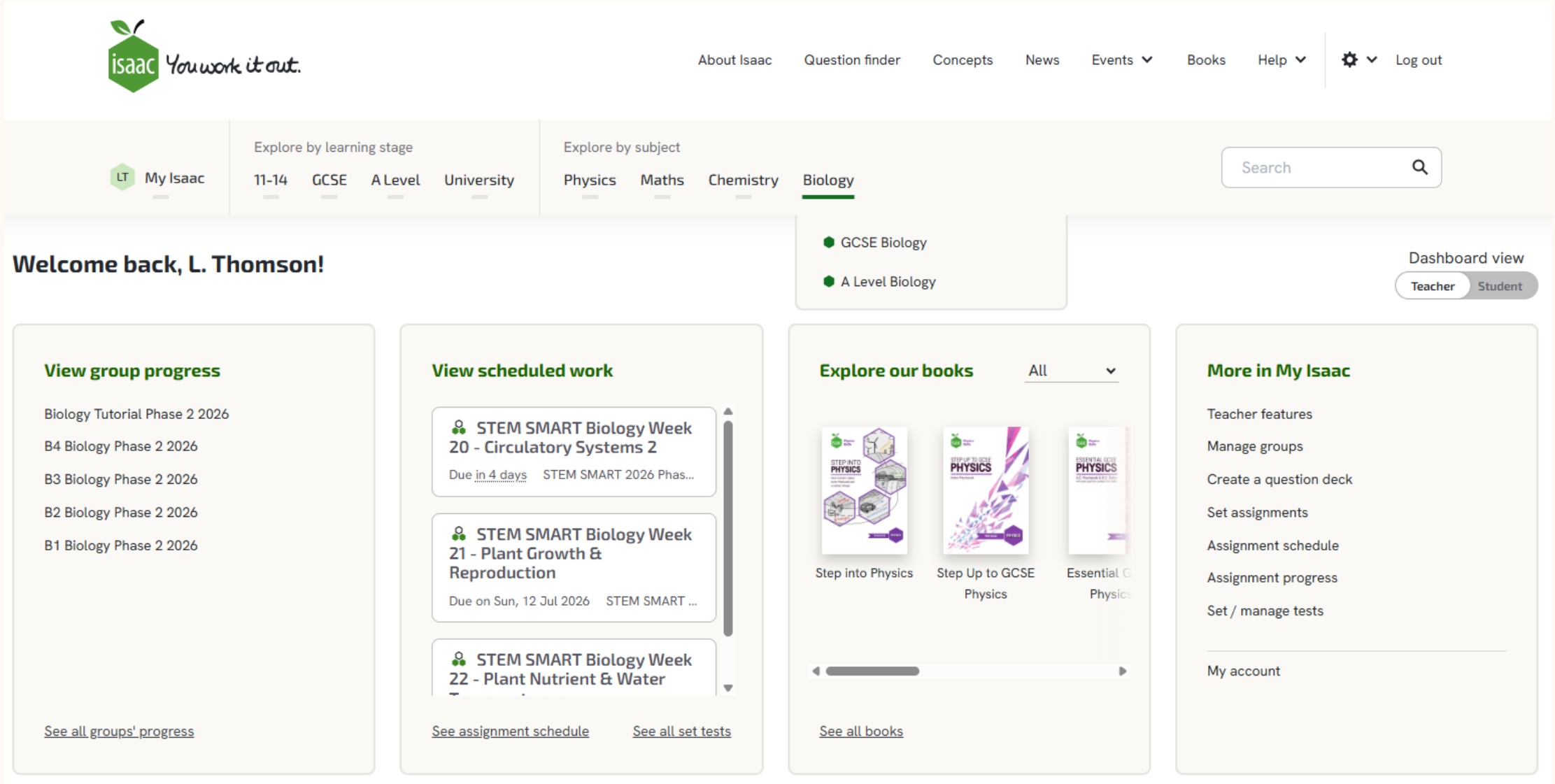
# Isaac Biology Resources

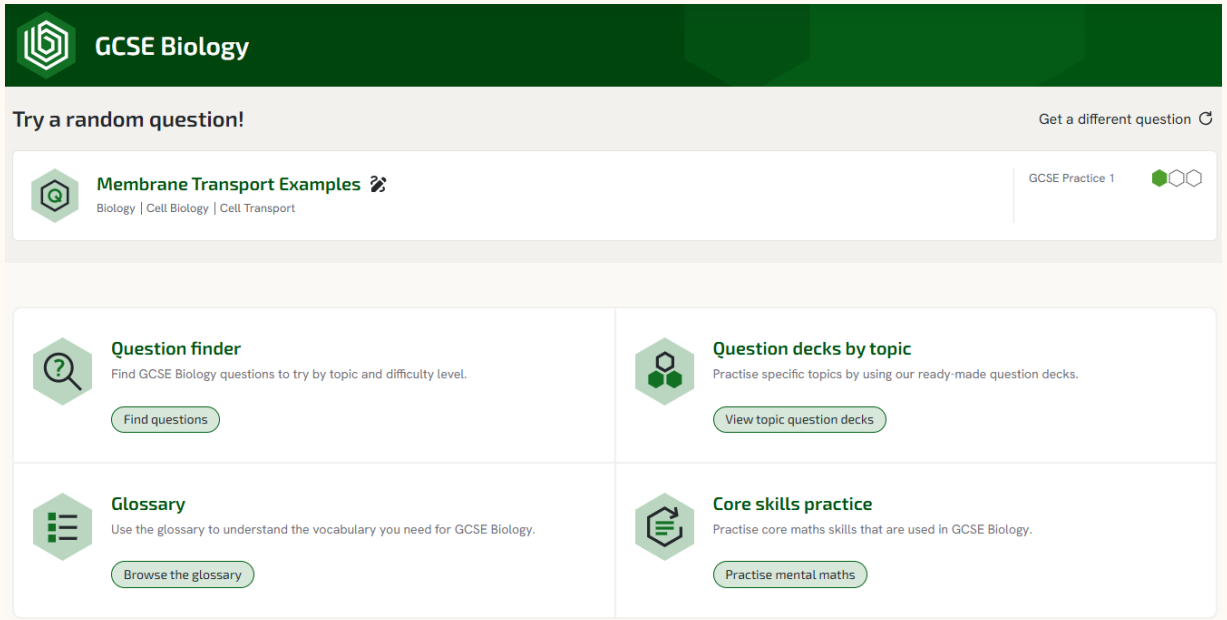
Lewis Thomson (lewis@isaacscience.org)

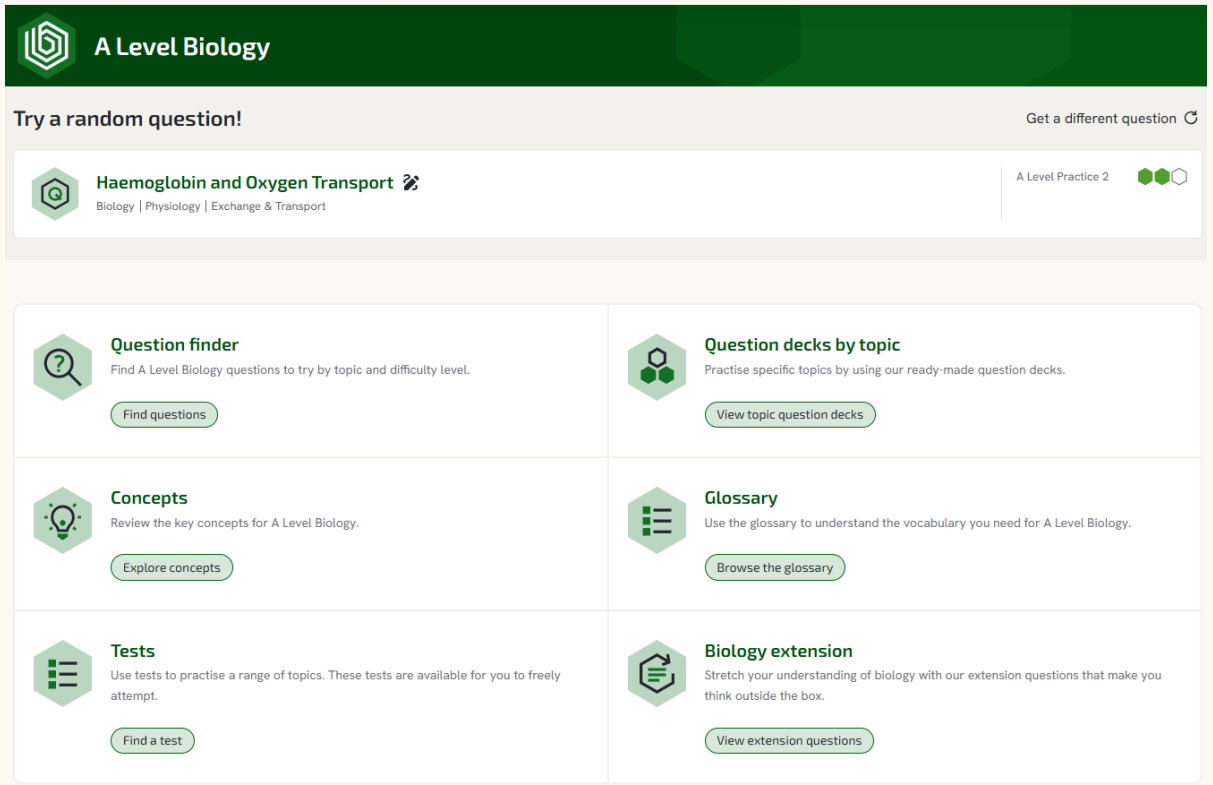
Bryony Yates (bryony@isaacscience.org)

# Overview of Isaac Biology resources

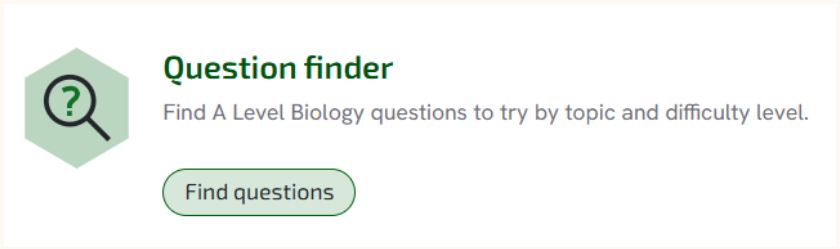


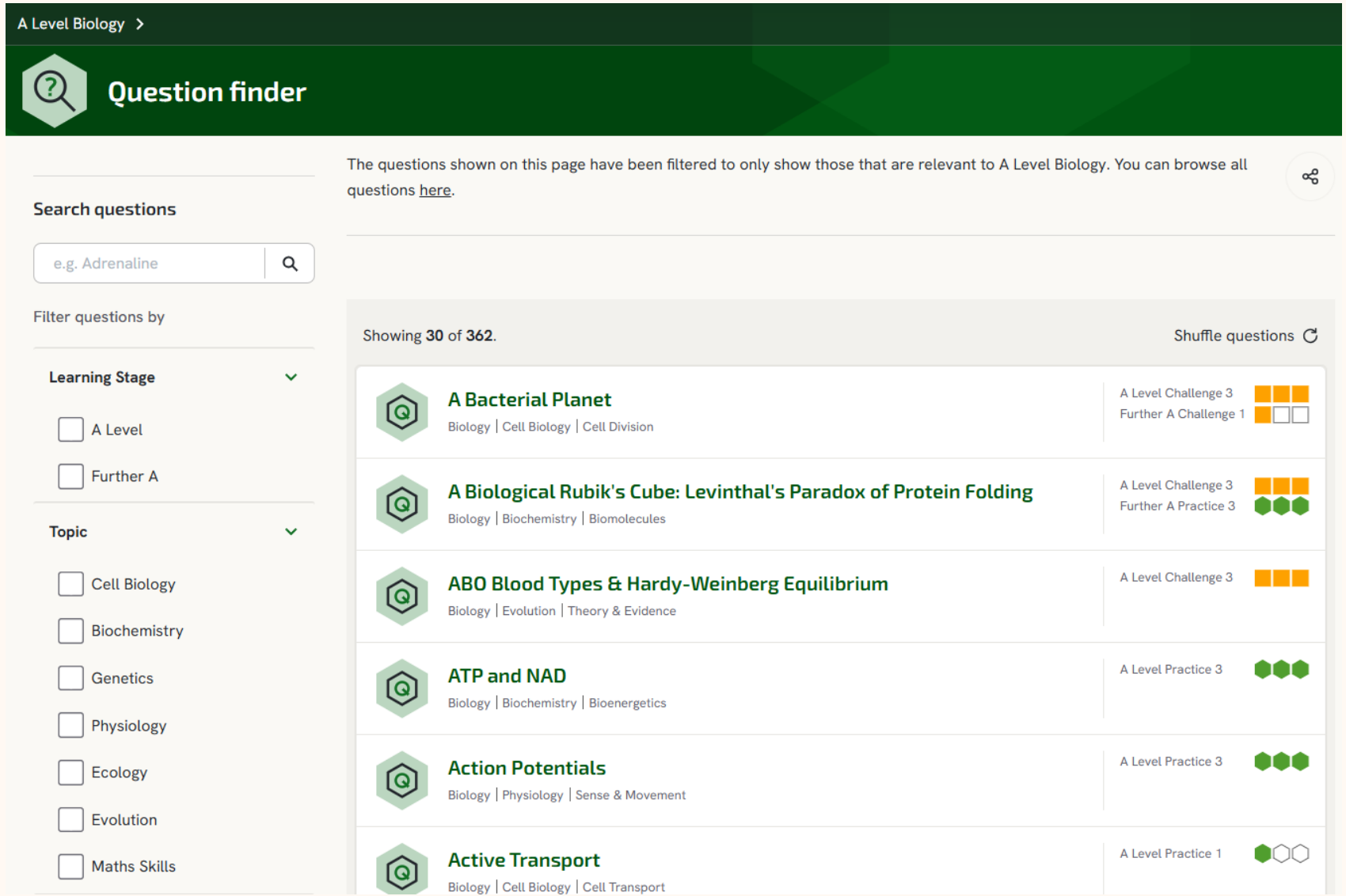
# Biology homepages



# Question finder

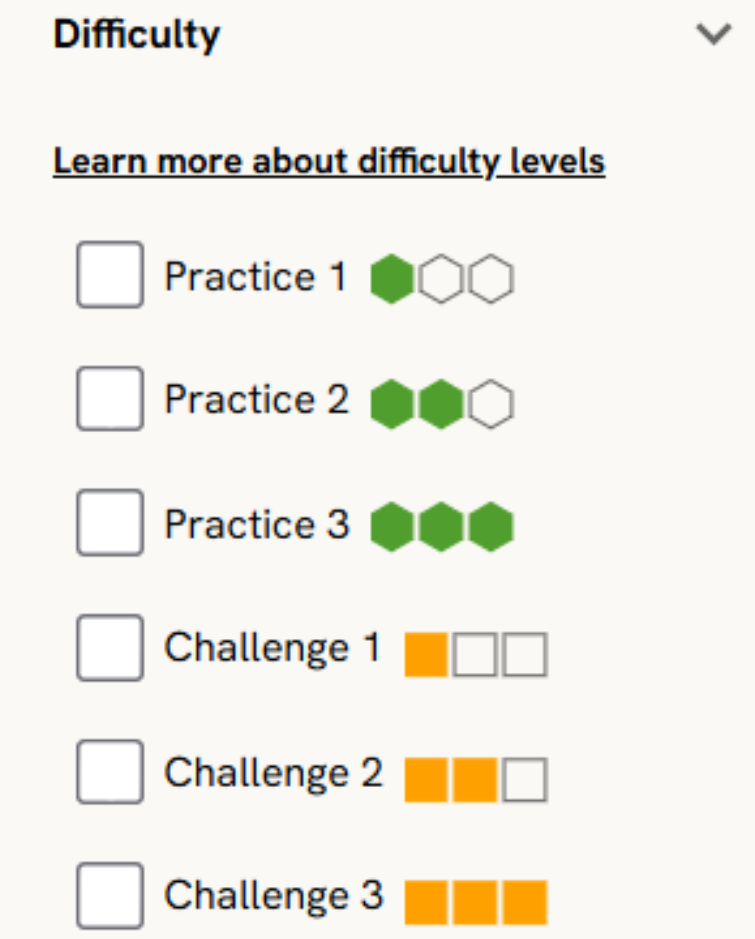


# Questions: fields & topics

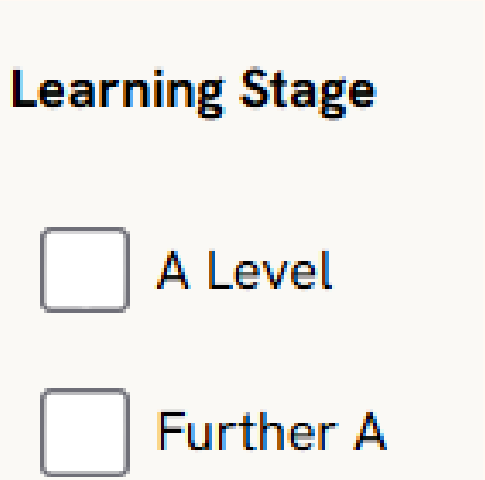
- [x] Cell Biology
  - [ ] Cell Structure
  - [ ] Cell Division
  - [ ] Cell Transport
  - [ ] Cell Specialisation
- [x] Biochemistry
  - [ ] Biomolecules
  - [ ] Enzymes
  - [ ] Bioenergetics
- [x] Genetics
  - [ ] Nucleic Acids
  - [ ] Protein Synthesis
  - [ ] Inheritance
  - [ ] Biotechnology
- [x] Physiology
  - [ ] Nutrition & Digestion
  - [ ] Exchange & Transport
  - [ ] Homeostasis & Response
  - [ ] Sense & Movement
  - [ ] Health & Disease
  - [ ] Reproduction & Development
- [x] Ecology
  - [ ] Populations
  - [ ] Ecosystems
  - [ ] Biodiversity
- [x] Evolution
  - [ ] Variation
  - [ ] Theory & Evidence
  - [ ] Speciation
  - [ ] Classification
- [x] Maths Skills
  - [ ] Statistical Tests

# Questions: difficulty




- Practice:
  - Recall of information, basic calculations
  - Common question types
    - Drag & drop
    - Short answer (1 or 2 word)
    - Reorder
- Challenge
  - Application, logic & reasoning, problem-solving
  - Common question types
    - Item (multiple choice with multiple correct options)
    - Calculation

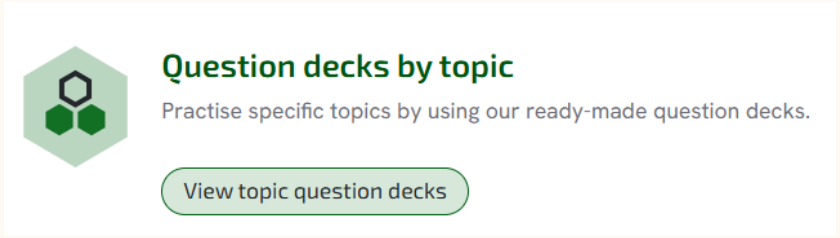
# Questions: learning stage

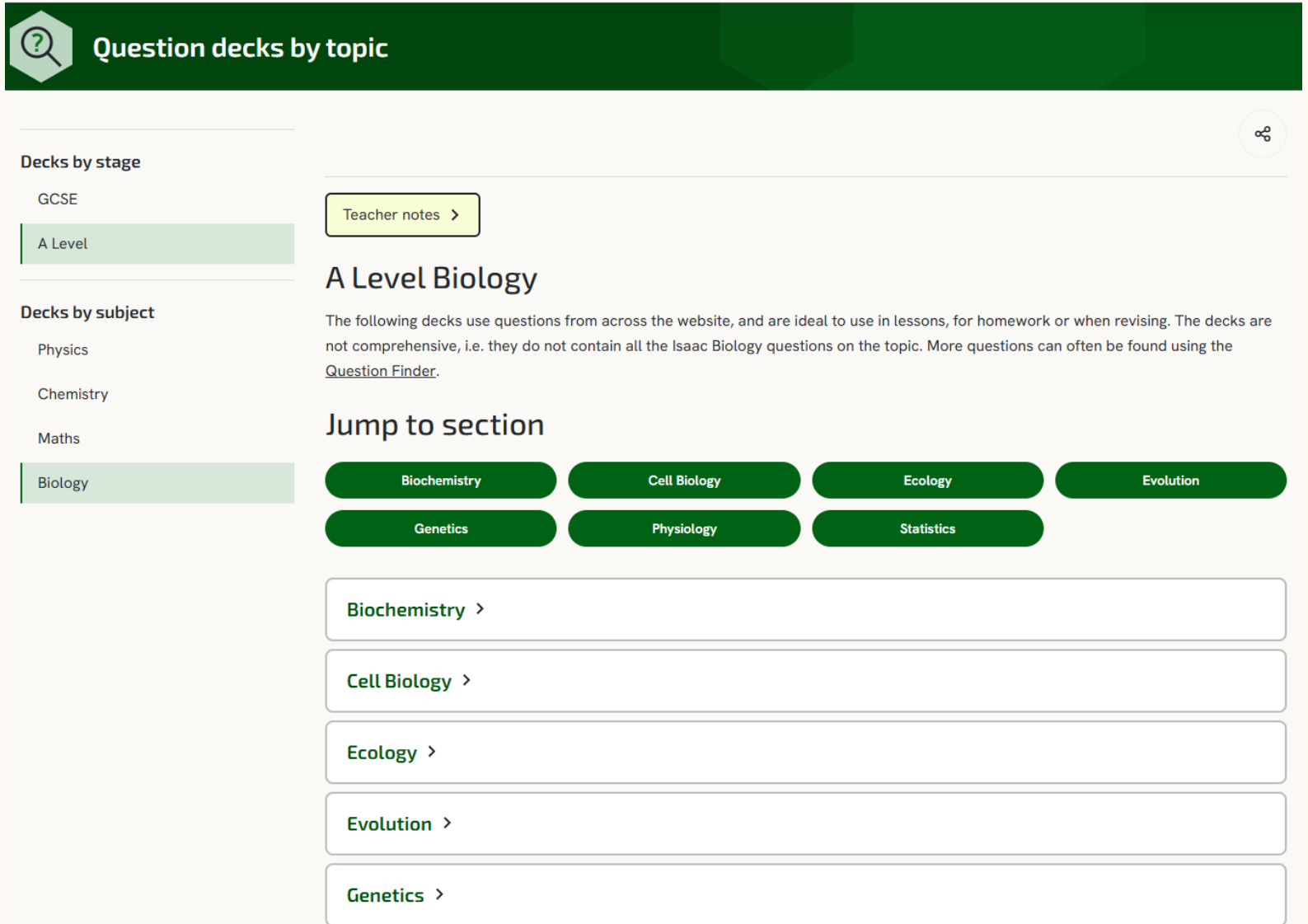



- More than just A Level biology required
- May require A Level maths or chemistry

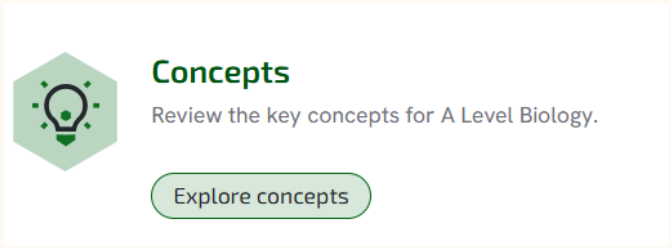
*This question involves using **exponentials and logarithms**, which are part of A Level Maths. For more information please check with your teacher.*

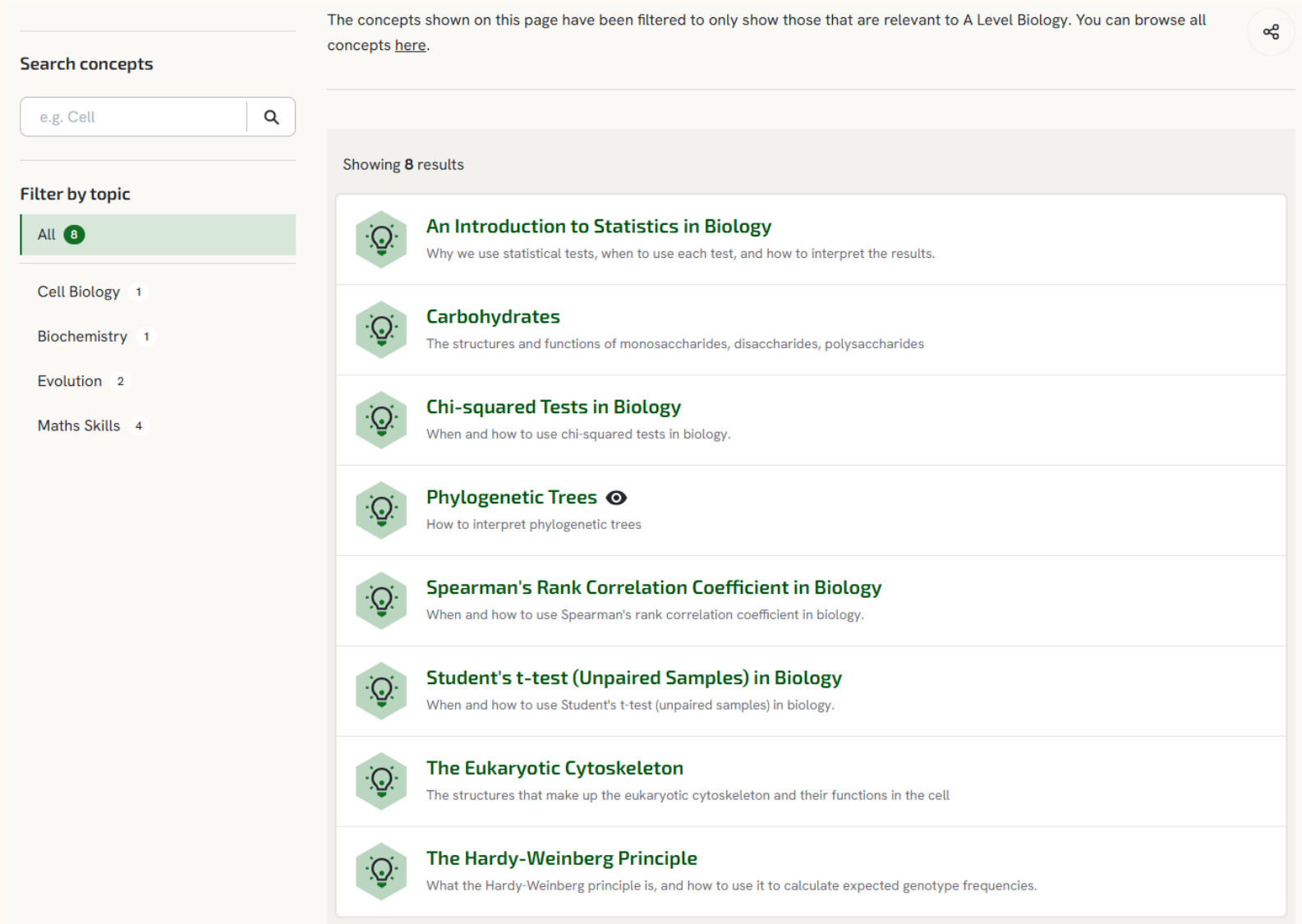
# Question decks by topic



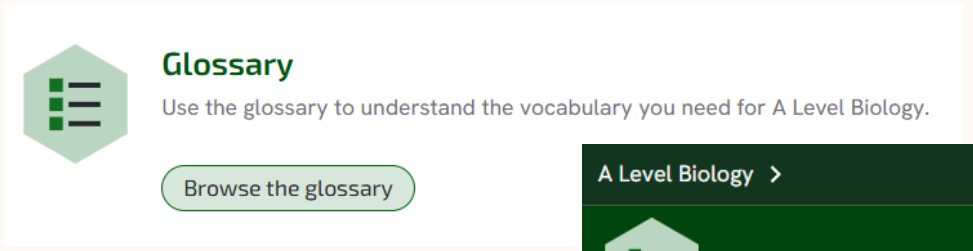



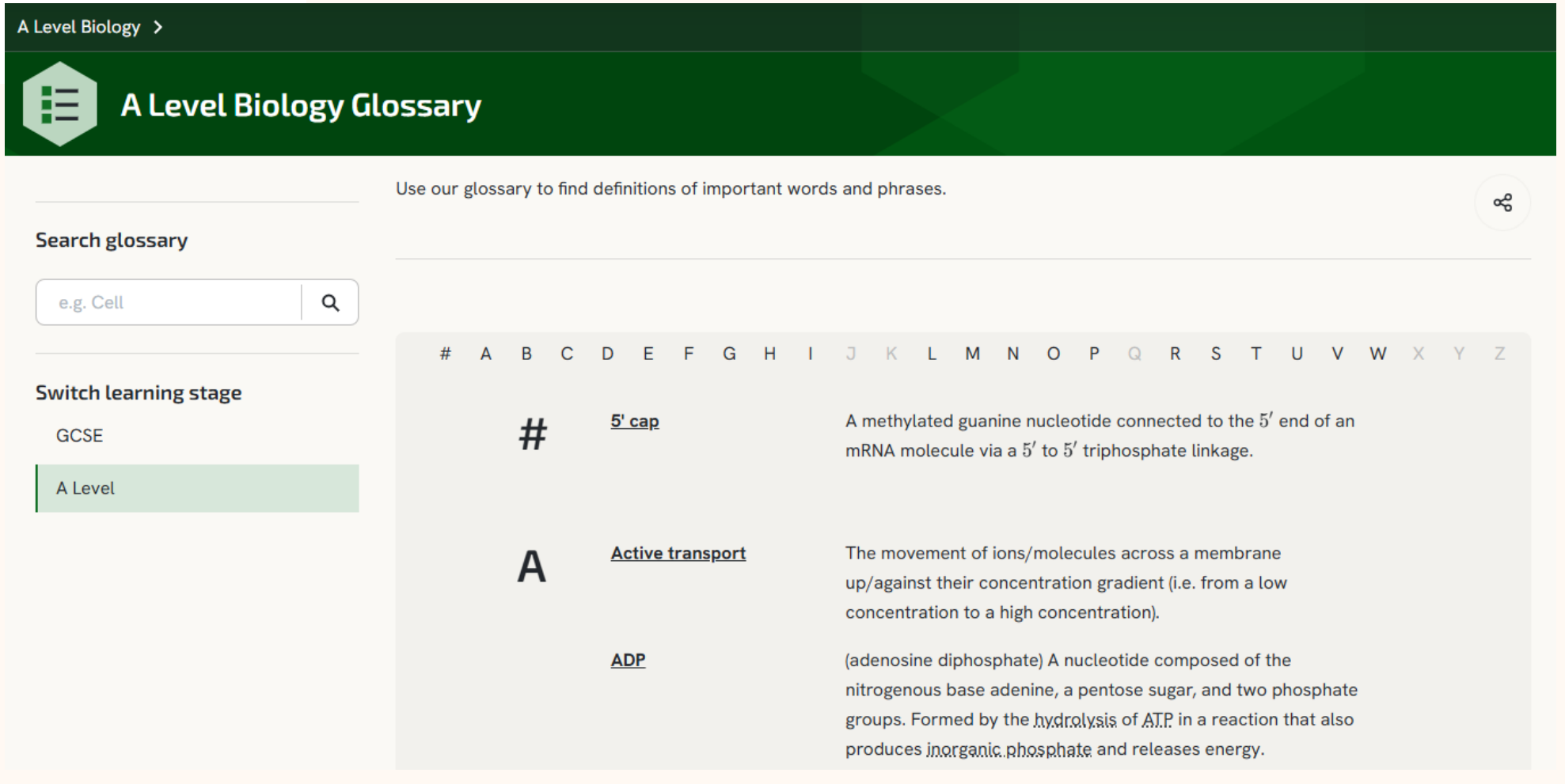
# Concept pages (A Level only)



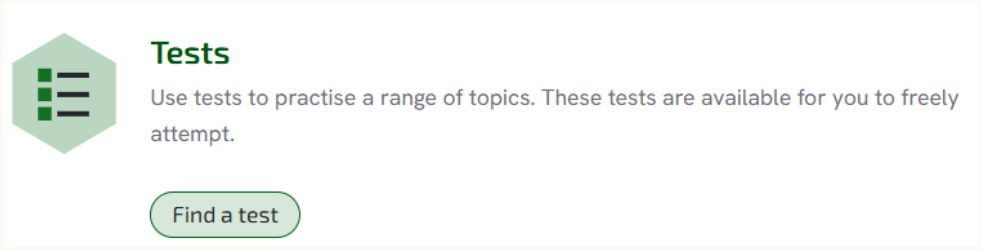



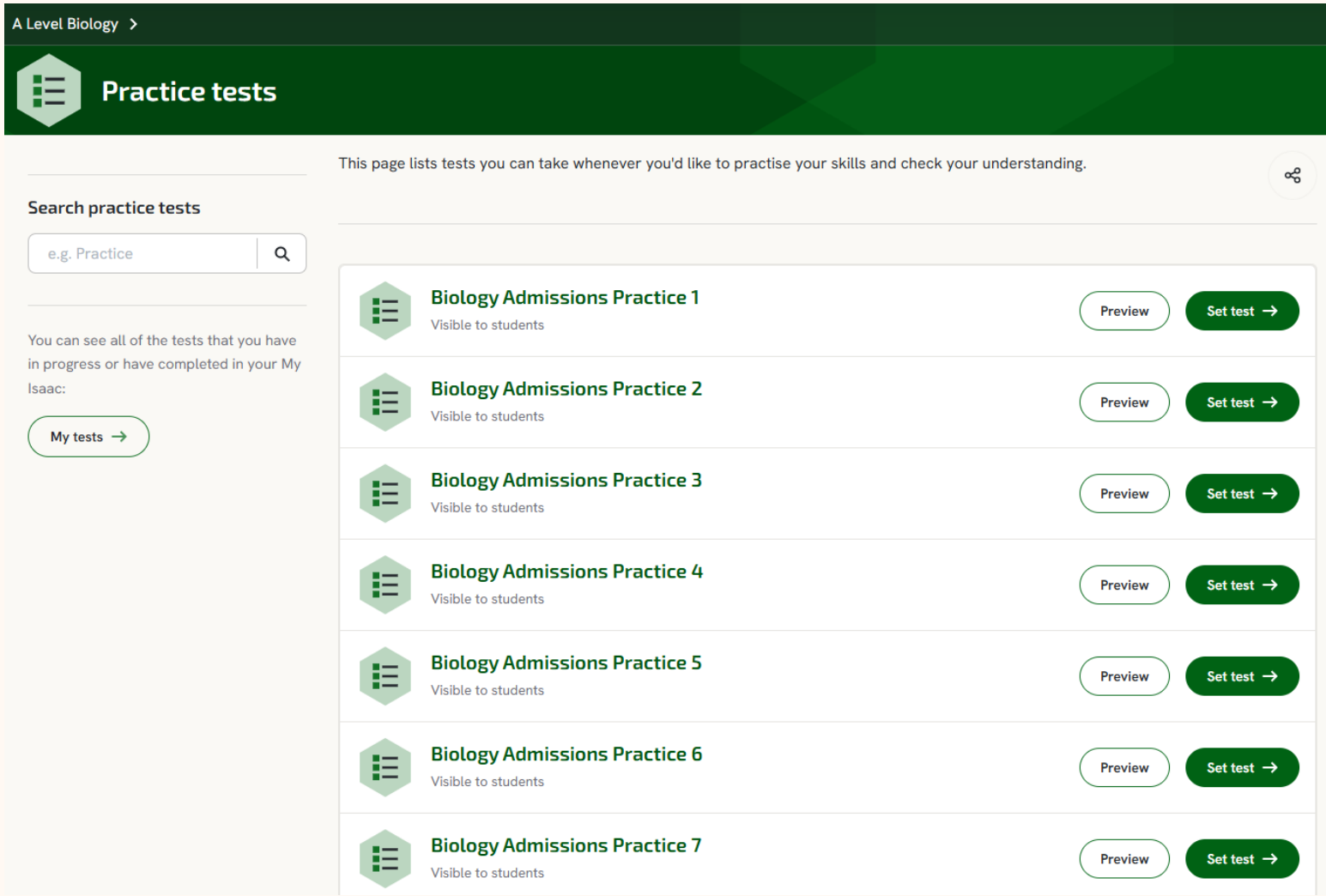
# Glossary


Glossary
Use the glossary to understand the vocabulary you need for A Level Biology.
Browse the glossary

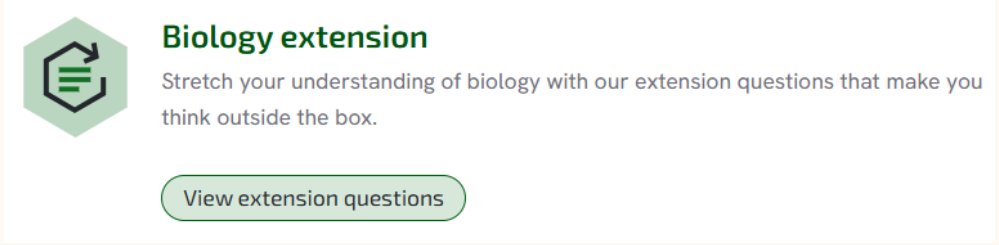

A Level Biology >
A Level Biology Glossary
Use our glossary to find definitions of important words and phrases.
Search glossary
e.g. Cell
Switch learning stage
GCSE
A Level
# A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
#
5' cap
A methylated guanine nucleotide connected to the 5′ end of an mRNA molecule via a 5′ to 5′ triphosphate linkage.
A
Active transport
The movement of ions/molecules across a membrane up/against their concentration gradient (i.e. from a low concentration to a high concentration).
ADP
(adenosine diphosphate) A nucleotide composed of the nitrogenous base adenine, a pentose sugar, and two phosphate groups. Formed by the hydrolysis of ATP in a reaction that also produces inorganic phosphate and releases energy.

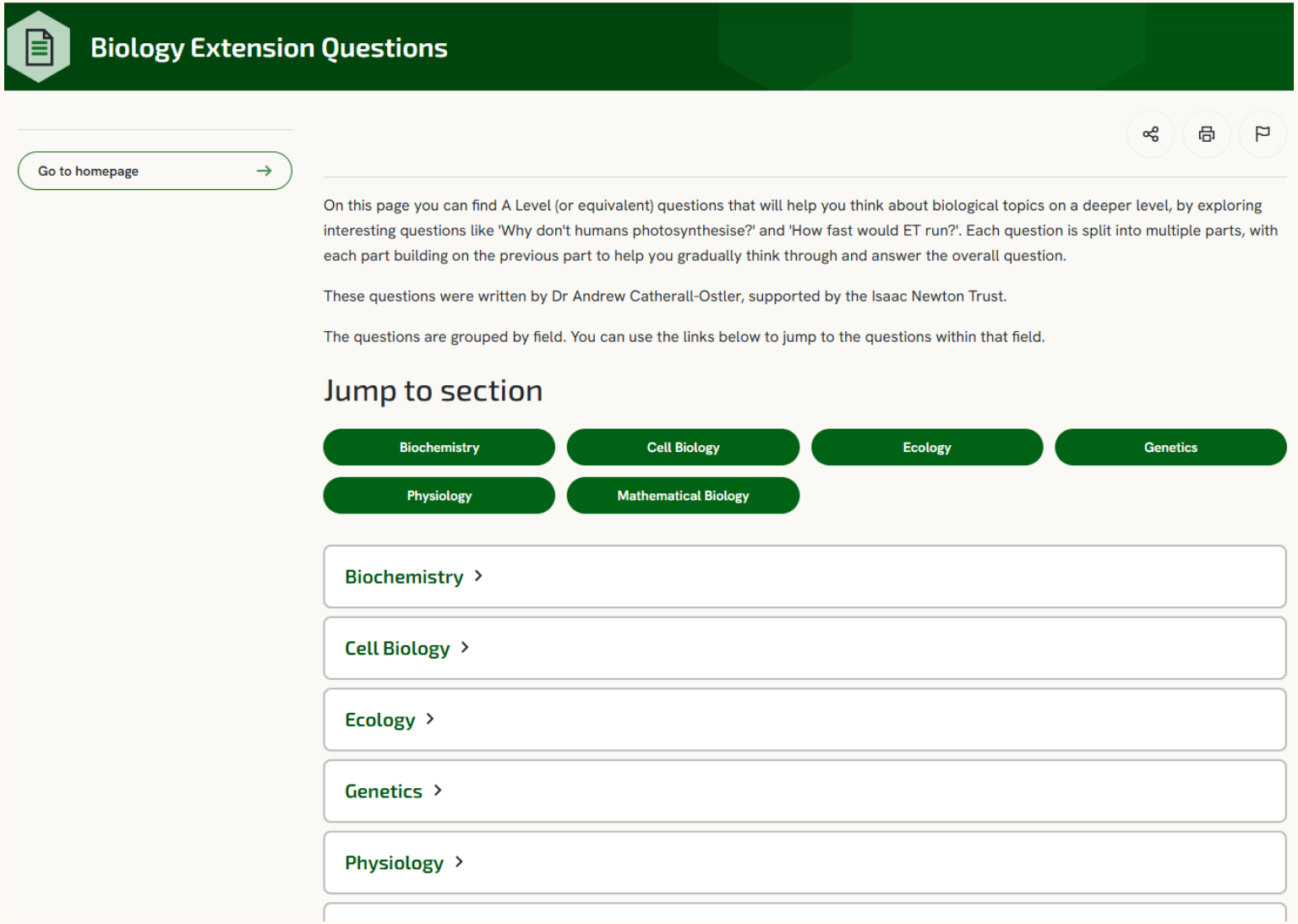
# Tests (A Level only)



# Extension questions (A Level only)



# Types of GCSE Practice Questions

## Recall

Part A

**Animal cell**

Label the organelles and structures in this diagram of an animal cell.

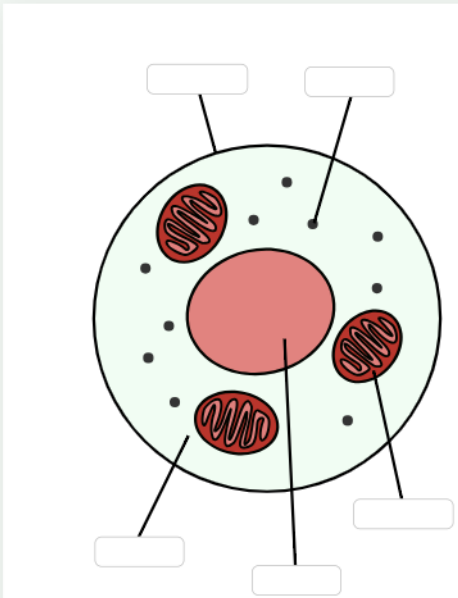


**Figure 1:** An animal cell.

Check my answer

## Diagnostic

Part A

**Nature**

Which of the following statements about enzymes are **true**? Select all that apply.

- [ ] Enzymes are microorganisms
- [ ] Enzymes are alive.
- [ ] Very high temperatures kill enzymes.
- [ ] Enzymes are proteins.
- [ ] Enzymes are cells.

Check my answer

## Investigative

A class of students investigated osmosis using pieces of carrot. They prepared four solutions, each containing a different concentration of sucrose (a type of sugar). They measured the initial mass of their pieces of carrot, then left one piece of carrot in each solution, or distilled water, for $24\,\text{hours}$. They removed the carrot pieces from the solutions, dried them and measured their mass again. They then calculated the change in mass (in g) and the percentage (%) change in mass.

Part A

**Completing the data table**

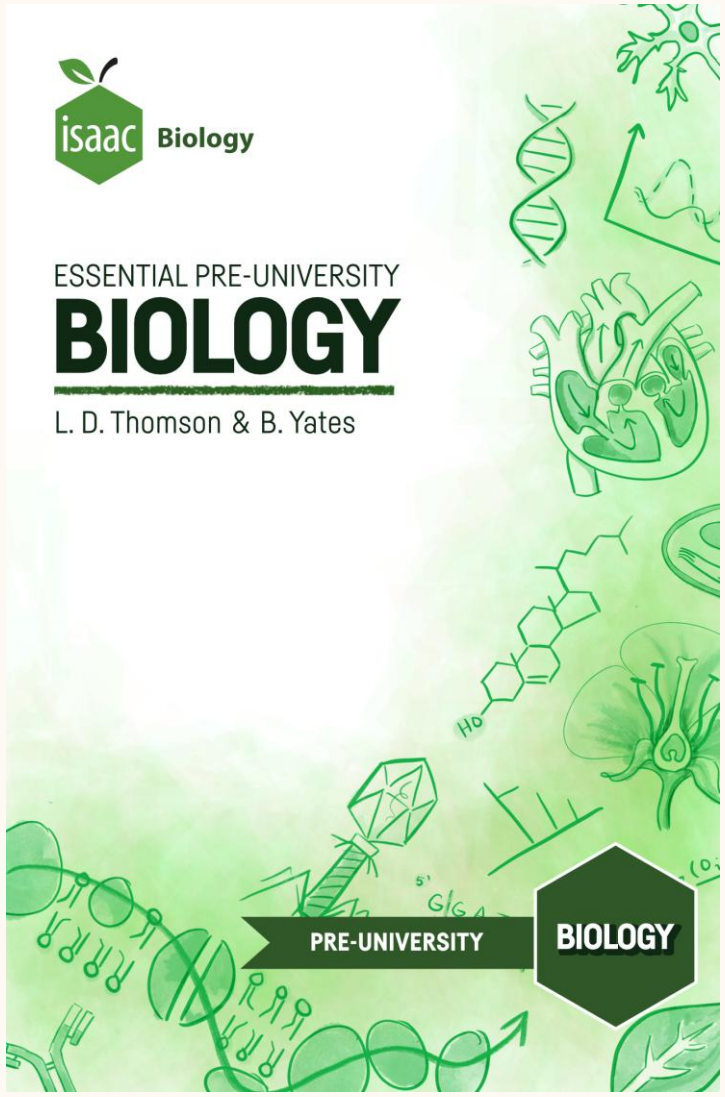
The table below contains some of the measurements from the experiment. Fill in the missing measurements. Round all values to 1 decimal place.

| Sucrose concentration ( $\text{mol/dm}^3$) | Initial mass ( g) | Final mass ( g) | Change in mass (g) | Change in mass (%) |
|---|---|---|---|---|
| 0.0 | 5.0 | 5.4 | 0.4 | 8.0 |
| 0.2 | 5.1 | 5.2 | | |
| 0.4 | 6.2 | | | −1.6 |
| 0.6 | | 5.7 | −0.2 | |
| 0.8 | 5.0 | 4.8 | | |

Check my answer

# Practice vs Challenge Comparison – Meiosis

# A Level biology book (coming soon)




- Compilation of (updated) A Level biology questions from the site
- Release early 2027
- 9 chapters
  - A–H: Biology chapters
  - I: Statistics
- Can be used standalone or in conjunction with the online book (free).

# A Level biology book

Part A
Name the cell

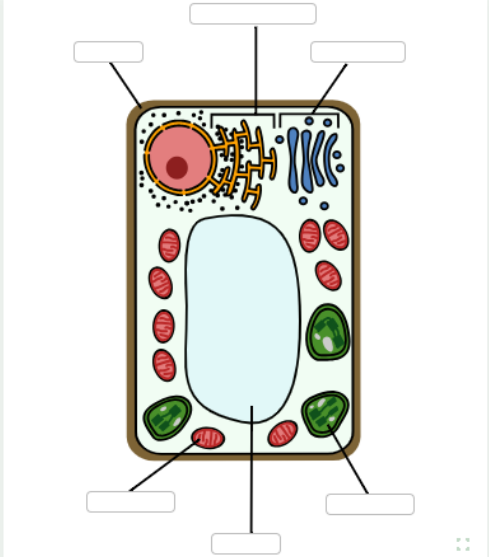
What kind of cell is shown above?

- Plant cell
- Animal cell
- Bacterial cell

Check my answer

Part B
Name the organelle

Label the organelles and structures in this diagram of a cell.



Check my answer

Part C
Structure C

What is structure C made of?

- chlorophyll
- phospholipids
- cellulose
- peptidoglycan
- chitin

Check my answer

Part D
Organelle E

What is the name of the membrane that surrounds organelle E?

- cell membrane
- capsule
- tonoplast
- inner membrane
- cisternae

Check my answer

Part E
Organelle F

What is the primary role of organelle F?

- photosynthesis
- aerobic respiration
- keeps the cell turgid
- digestion of ingested material

Check my answer

## B1.3 Identifying Organelles 1

A Level P1

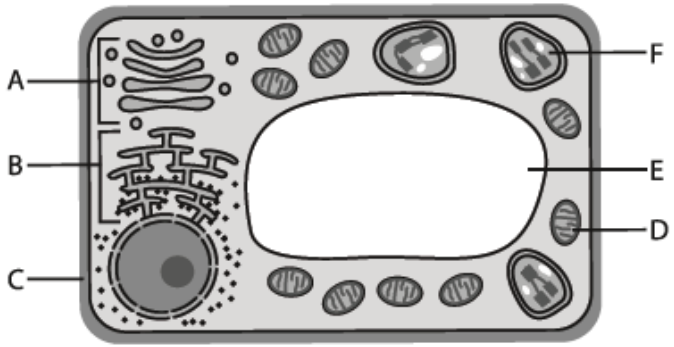



**Figure 31** A diagram of a cell.

a) What kind of cell is shown above? Write down the correct answer from the list below.

- Animal cell
- Plant cell
- Bacterial cell

b) Match the labels in Figure 31 to the organelles in the list below.

cell wall, centriole, chloroplast, endoplasmic reticulum, flagellum, Golgi apparatus, mitochondrion, vacuole

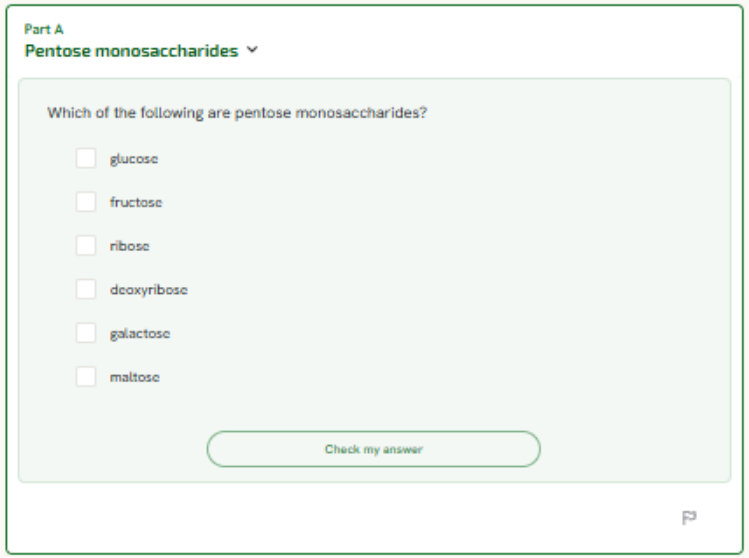
c) What is organelle C made of?

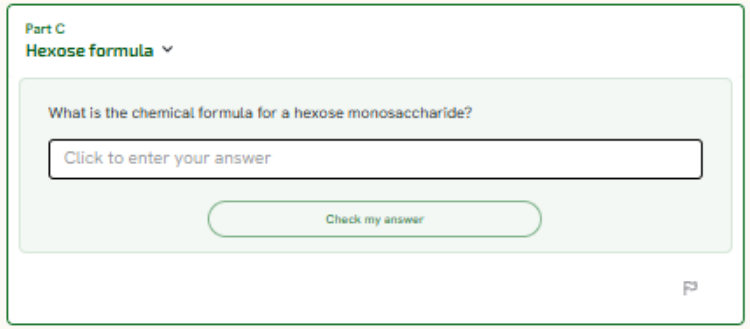
d) What is the name of the membrane that surrounds organelle E? Write down the correct answer from the list below.

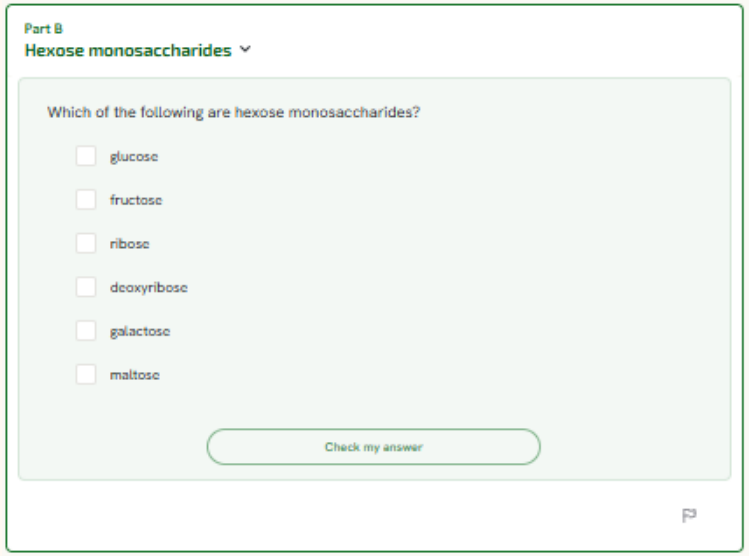
- capsule
- cell membrane
- cisternae
- inner membrane
- tonoplast

e) What is the primary role of organelle F?

# A Level biology book




## A1.1 Monosaccharides

A Level P1

Monosaccharides are the smallest carbohydrates, and are the monomers from which larger carbohydrates are made. Monosaccharides are categorised based on how many carbon atoms they contain, as follows:

- **triose:** three carbons
- **tetrose:** four carbons
- **pentose:** five carbons
- **hexose:** six carbons
- **heptose:** seven carbons

a) Name two pentose monosaccharides.
b) Name three hexose monosaccharides.
c) Write down the chemical formula for a hexose monosaccharide.

# A Level biology book

Part A

Cysteine and valine

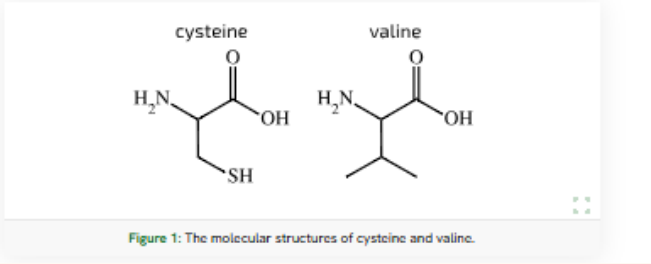
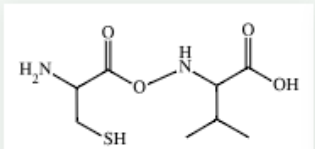
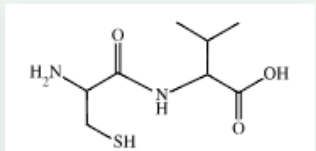



Figure 1: The molecular structures of cysteine and valine.

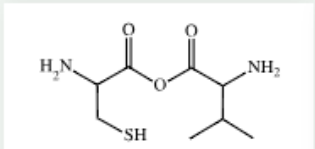
**Figure 1** shows the molecular structures of two amino acids: cysteine and valine. Which of the molecules below are dipeptides that could be produced by a condensation reaction between cysteine and valine? Select all that apply.



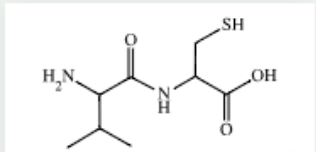
A



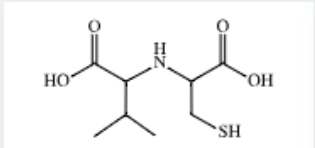
B



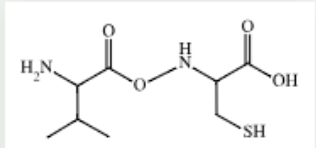
C



D



E



F

A3.2 Identifying Dipeptides

A Level P1

a) Draw two dipeptides that could be produced by a condensation reaction between the amino acids cystine and valine, shown below in Figure 13.

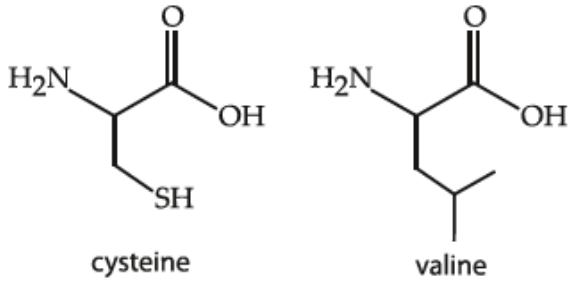


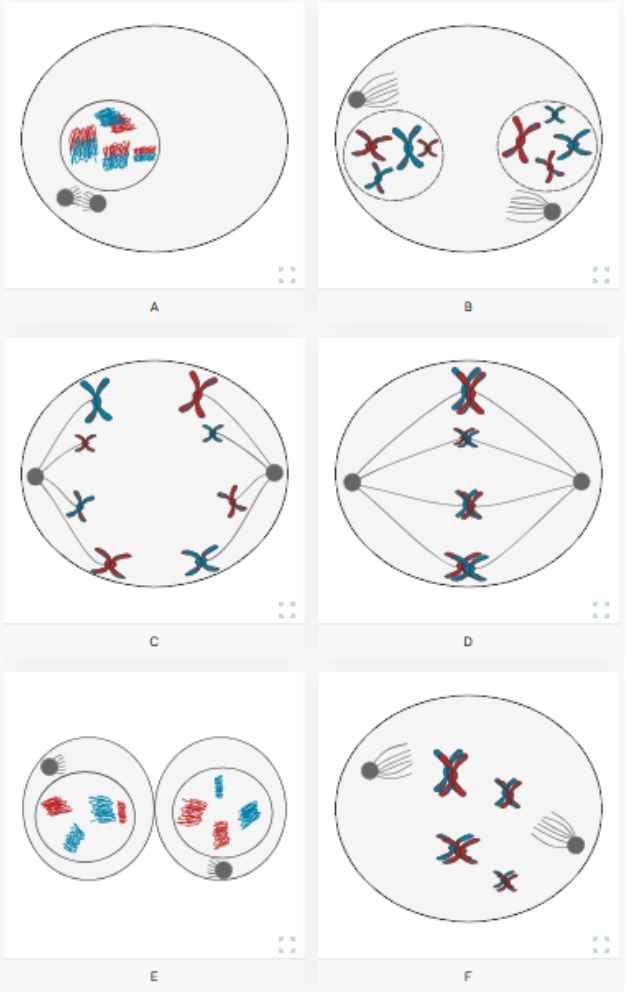

**Figure 13**

# A Level biology book

Part A

Meiosis I

The images below represent different stages of meiosis I, for an organism with a diploid chromosome number of 8.




Put the stages above in the correct order, and match the name to each stage.

| Order | Stage |
|---|---|
| A | interphase |
| | |
| | |
| | |
| | |
| E | cytokinesis |

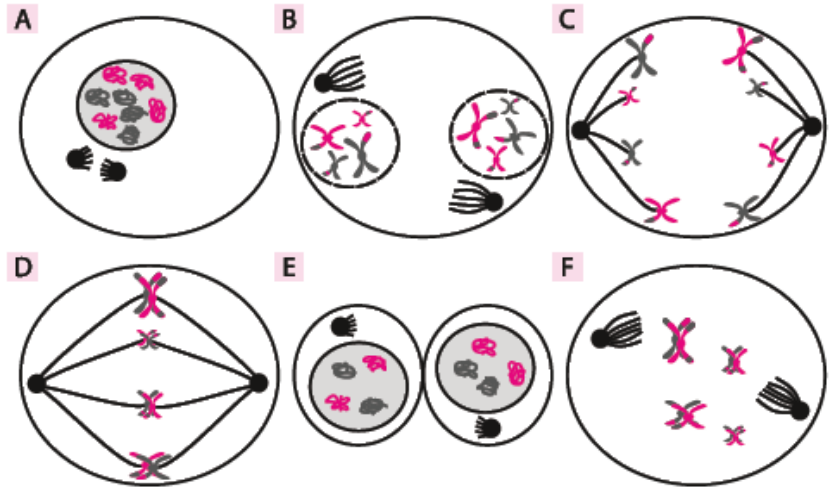
Items:

B C D F anaphase I metaphase I prophase I telophase I

## B2.6 Stages of Meiosis

A Level P1

a) The images below represent different stages of meiosis I, for an organism with a diploid chromosome number of 8.




Redraw and fill in the table below to put the stages above in the correct order, and write the name for each stage.

| Image | Stage |
|---|---|
| A | interphase |
| | |
| | |
| | |
| | |
| E | cytokinesis |

# Isaac Biology Resources

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Bryony Yates (bryony@isaacscience.org)